

1. High-Level Document (HLD) (Expanded)

Project Name:

Amazon Sales Data Analysis Dashboard

The project is focused on visualizing and analyzing Amazon sales data through a Power BI dashboard. The primary goal is to create a centralized platform for stakeholders to access detailed reports about Amazon's sales performance, allowing them to make data-driven decisions. This dashboard will help users understand sales trends, revenue generation, and other business insights.

Your Name:

Prince Singh

You will take responsibility for the project, and this section provides the necessary context regarding your contribution.

Date:

February 11, 2025

This is the date on which the report is being written, ensuring that stakeholders are aware of the document's freshness.

Abstract:

The **Abstract** section is a brief but comprehensive summary of the entire project. It should convey the high-level goals, the tools used, and the key functionalities of the dashboard.

In this project, you'll be utilizing Power BI to visualize Amazon sales data in an interactive and insightful way. The dashboard is structured to display a range of KPIs including monthly revenue, monthly profit, monthly cost, and revenue by item type, region, and sales channel. By incorporating slicers for dynamic filtering, the dashboard allows users to customize the data views based on their needs. The purpose of this dashboard is to help stakeholders monitor sales performance, track key metrics, and identify areas that may need strategic adjustments. The dashboard is designed to be visually engaging while providing detailed insights into revenue, profit, and cost distribution.

Introduction:

The **Introduction** provides a comprehensive overview of the project and its purpose, including the technology and platform used.

Power BI is a powerful business analytics tool from Microsoft that allows users to visualize data and share insights. In this project, Power BI is used to create a dynamic dashboard for visualizing Amazon's sales data. The dashboard includes key performance indicators (KPIs) such as monthly revenue, profit, and cost. The purpose of this dashboard is to provide a high-level overview of Amazon's sales performance and allow users to drill down into the data for deeper insights. By

interacting with slicers and filters, users can see how sales metrics vary by region, sales channel, order priority, and other dimensions.

Why We Use HLD:

A **High-Level Document (HLD)** is essential because it serves as the foundation for the entire project. It communicates the system's architecture and high-level design, helping the team understand the project's goals and approach before diving into the detailed technical aspects. HLD sets clear expectations for stakeholders, business analysts, and developers by presenting an overview of the project structure, its components, and how it will operate. For your project, the HLD will help your team and supervisors quickly understand the purpose of the Power BI dashboard, its high-level architecture, and the metrics it will track.

General Description:

The **General Description** should give a detailed explanation of the project, breaking down the purpose, features, and layout of the dashboard.

The Amazon Sales Data Dashboard is designed to provide stakeholders with an in-depth understanding of Amazon's sales performance over time. The dashboard aggregates data and visualizes it using various charts, graphs, and tables. The primary KPIs displayed include **monthly revenue**, **monthly profit**, **monthly cost**, and revenue segmented by **item type**, **order priority**, **region**, and **sales channel**. The dashboard allows users to interact with the data via slicers, providing options to filter the data based on month, region, and year. By doing so, users can gain insights into sales trends, profitability, and costs across different geographies, product categories, and sales channels.

This dashboard is intended to streamline decision-making and provide a quick, accessible overview of Amazon's sales performance at a glance. Additionally, it helps identify specific trends that may indicate areas where improvement is needed, such as underperforming regions or product categories.

Design Details:

The **Design Details** section focuses on the practical aspects of how the dashboard is structured.

- **User Interface (UI):** The dashboard is designed to be visually appealing and user-friendly. KPIs are displayed at the top of the dashboard to give an overview of key metrics, while more detailed data is presented in visualizations below. The layout is organized, with graphs on the left-hand side and tables and slicers on the right-hand side. This ensures that users can quickly see key insights without feeling overwhelmed by information.
- **Data Model:** The data model in Power BI is created by importing the raw data and defining relationships between different tables. For instance, sales data is related to item types, regions, and order priorities, allowing users to drill down into different subsets of data. The model uses **star schema** or **snowflake schema** to ensure that the relationships between fact and dimension tables are logically structured, which enables efficient querying.

- **DAX Measures:** DAX (Data Analysis Expressions) is used to write the measures that calculate metrics like revenue, profit, and cost. These formulas are built to be dynamic, responding to slicer selections, and updating based on user input.
 - **Interactivity:** The interactivity of the dashboard is enhanced through the use of slicers. Users can filter the data based on specific regions, months, or years. Additionally, drill-through functionality allows users to click on any data point (e.g., a specific region or item type) and view more detailed information in a separate report page.
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KPIs:

Key Performance Indicators (KPIs) are crucial metrics that give insights into the overall performance of the business.

- **Monthly Revenue:** This metric tracks the total revenue generated by Amazon sales each month. It's important for understanding revenue trends and how sales are performing over time.
- **Monthly Profit:** Profit is calculated by subtracting costs from revenue. Tracking monthly profit helps to evaluate how efficiently Amazon is managing its sales processes.
- **Monthly Cost:** This shows the total expenses incurred in producing or acquiring products. It's vital to track this to maintain healthy profit margins.
- **Total Revenue by Item Type:** This allows you to see how much revenue comes from each product category. It can help identify which product categories are driving the most sales.
- **Total Revenue by Order Priority:** This shows how revenue is distributed across different order priorities (e.g., high priority, medium priority, low priority). Understanding this distribution can help Amazon optimize its operations and prioritize orders accordingly.
- **Total Revenue by Region:** This metric breaks down revenue by geographical region, enabling stakeholders to see how different regions contribute to the overall revenue.
- **Total Revenue, Profit, and Cost by Year:** This provides year-over-year comparisons of revenue, profit, and costs. It helps assess how well the business is performing in the long term.